

Caio Jordão de Lima Carvalho

DATA ANALYST / DATA ENGINEER / ARTIFICIAL INTELLIGENCE RESEARCHER

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Summary

Current MSc. student at Federal University of Bahia and Artificial Intelligence Researcher at Supercomputing Center for Industrial Innovation - SENAI CIMATEC. 3+ years experience specializing in the back-end/data engines development, information retrieval, and machine learning. Super nerd, easy learner, Linux lover and KDE contributor. Problem solver, interested in new challenges, and developing new tools.

Work Experience

Supercomputing Center for Industrial Innovation - SENAI CIMATEC

Salvador, Bahia, Brazil

ARTIFICIAL INTELLIGENCE RESEARCHER

Dec. 2019 - Present

- Working on a project involving the development of machine learning methods for predictive maintenance.
- Implemented and evaluated clustering approaches (KNN, RRCF, Bootstrap, Luminol) for anomaly detection.
- Built pipelines focused on exploratory data analysis, including data cleaning, filtering, normalization and statistical evaluation.
- Implemented deep learning methods (LSTM, GRU, MLP, CNN) for time-series prediction.

KDE Community

Remote

SOFTWARE DEVELOPER & CONTRIBUTOR

Dec. 2017 - Present

- KDE e.V. member.
- Maintainer of markK, a dataset annotation tool.
- Improved SMART support and KAuth operation flow in KDE Partition Manager during Season of KDE 2018.
- Implemented LVM Volume Group support in Calamares and MDRAID support in KDE Partition Manager during Google Summer of Code 2018.
- Mentored students during Google Summer of Code 2019, Google Code-in 2018 and Season of KDE 2020.
- Technologies used: C++, C, Qt (widgets), CMake, SMART (libatasmart, smartctl), DBus, mdadm, dmraid, lvm and Linux partitioning tools.

Escavador

Salvador, Bahia, Brazil

BACK-END DEVELOPER

Oct. 2018 - Nov. 2019

- Designed web crawler architectures for systems focused on information retrieval of Brazilian legal cases and laws, which extracts millions of documents from different courts from Brazil.
- Integrated search engines and implemented several API microservices using Python, asynchronous libraries and MySQL.
- Implemented artificial intelligence models focused on information filtering and natural language processing.
- Built fully automated CI pipelines on Gitlab CI for containerized applications using Docker.

Labrasoft - IFBA

Salvador, Bahia, Brazil

UNDERGRADUATE RESEARCH ASSISTANT

Aug. 2017 - Aug. 2019

- Developed and analyzed stock market bots based on Multilayer Perceptrons and Recurrent Neural Networks (LSTM, GRU) using MQL5, MetaTrader and Python (Keras, Tensorflow).
- Deployed object tracking algorithms based on Deep Learning approaches in embedded systems using Intel Movidius, Raspberry Pi and Python.
- Implemented a generative-adversarial model for image enhancement during Minimally Invasive Surgeries.

TecnoTRENDS

Salvador, Bahia, Brazil

SOFTWARE DEVELOPMENT INTERN

Oct. 2016 - Oct. 2017

- Developer of a module for an educational ERP system focused on elementary schools using C#, ASP.NET and SQL Server.
- Released versions from this module for all the elementary schools from the city of Feira de Santana, Bahia.
- Development of new features, graphical interfaces and SQL routines.

Education

Federal University of Bahia

Salvador, Brazil

MASTER OF SCIENCE IN COMPUTER SCIENCE

Feb. 2020 - Jun. 2022

- Fields of study: Data Science, Artificial Intelligence, Statistics.

IFBA (Federal Institute of Science, Education and Technology of Bahia)

Salvador, Brazil

B.TECH. IN SYSTEMS ANALYSIS AND DEVELOPMENT

Jun. 2016 - Oct. 2019

- GPA: 8.6 / 10.0
- Graduation Thesis Title: "Deep Autoencoder Approach to Enhance Low-Light Images during Minimally Invasive Procedures".
- Relevant Coursework: Data Structures, Algorithms, Software Architecture, Design Patterns, Distributed Systems, Software Engineering, Object-Oriented Design, Software Project Management.
- Technical Skills: C++ (3yrs), C, Python, Java, Git, Django, Qt, SQL.
- Student representative of this program while studying at IFBA.

- GPA: A-
- Relevant Coursework: Statistics, Probability, Mathematics.
- Selected for a scholarship to the College of the Rockies by the Emerging Leaders in the Americas Program.

Honors & Awards

2018	Grand Prize Winner , Season of KDE 2018	Remote
2018	Winner , Innovative Challenge - IFBA Projects' Hotel	Salvador, Brazil
2017	Gold Medalist , ProgBASE 2017 - Competitive Programming Contest	C. das Almas, Brazil
2018	Bronze Medalist , 2nd Programming Contest of Bahia	Salvador, Brazil

Relevant Presentations

Akademy 2019

Milan, Italy

PRESENTER FOR <KPMCORE IMPLEMENTATIONS AND GOALS FOR THE FUTURE>

September, 2019

- Talked about my work on kpmcore and described some updates and features from KDE Partition Manager 4.0.

QtCon Brazil 2018

São Paulo, Brazil

PRESENTER FOR <COMPUTER VISION WITH QT QUICK AND OPENCV>

Nov. 2018

- Introduced how to integrate the OpenCV library with Qt Quick in mobile environments.

V Workshop Labrasoft-GPETEC

Salvador, Brazil

PRESENTER FOR <INTRODUCTION TO KERAS AND DEEP LEARNING MODELS>

Sep. 2018

- Explained about the Keras library, including an introduction to deep learning models, such as Convolutional Neural Networks, Autoencoders and Generative Adversarial Networks.

ERBASE - Encontro Regional de Computação Bahia-Alagoas-Sergipe

Aracaju, Brazil

PRESENTER FOR <ALPHA B3 - EXPERT ADVISOR USING ARTIFICIAL NEURAL NETWORKS AND GENETIC ALGORITHMS>

Aug. 2018

- Explained about the AlphaB3 project, which is an expert advisor to predict stock market trends based on neural networks and genetic algorithms.

Publications

AlphaB3 - Expert Advisor using Neural Networks and Genetic Algorithms to Predict Stock Market Trends

ERBASE 2018

CO-AUTHOR

Aug. 2018

- Paper published in ERBASE 2018. Explained about AlphaB3 and its results on the prediction of the Brazilian stock market.

Organizer

2020	Organizer , Season of KDE 2020	Remote
2019	Organizer , LaKademy 2019	Salvador, Brazil
2018	Organizer , WebMedia 2018	Salvador, Brazil
2018	Organizer , ADS-Integra & ADS-Gênero	Salvador, Brazil
2017	Problem Writer , IFBA Competitive Programming Contest	Salvador, Brazil

Extracurricular Activity

ProgIFBA - Competitive Programming Group at IFBA

Salvador, Brazil

MEMBER

Dec. 2016 - Jan. 2019

- Created a group focused on competitive programming studies at my university.
- Proposed some training sessions and participated in some contests that have happened in my city.

Sonata Project

Salvador, Brazil

MEMBER

Feb. 2018 - Apr. 2019

- Development of a mobile platform to help people on learning how to read music scores.
- Worked with a team to submit this project to a innovative challenge at IFBA. Our project was the winner in this competition.